

Cybersecurity: Fundamental of cryptography

Cryptography (*secret writing*): the process of transforming/scramble data/information so that it cannot be understood by unauthorized users.

Steganography (*Greek: covered/hidden writing*): the process of hiding information—a covered data is readable if it is uncovered—whereas encrypted data can be understood **ONLY** by its intended recipient.

Please provide a **short and concise** definition/explanation for each of the cybersecurity terminologies. Each definition/explanation entry is worth **5** points.

Direction & Submission: use the downloaded document, provide the definition/explanation for each indicated row below. At the completion, submit the completed document through Canvas. You have up to **two weeks** to submit the assignment after the due date. Please note that there is a daily 3% deduction if submitted after the due date.

	Terms	Definition/explanation
1	Transposition	a data encryption technique that rearranges the characters or bits of a message according to a specific pattern to obscure its original meaning.
2	Substitution method	an encryption technique where elements of plaintext, such as letters or digits, are replaced with different elements according to a defined system to obscure the original data.
3	· Rotation13 (ROT13)	substitution cipher that shifts each letter in the alphabet by 13 places, making it a basic encryption method used for obfuscation rather than secure encryption.
4	· XOR	is a logical operation used in encryption where two binary inputs are compared, and the output is 1 if the inputs differ and 0 if they are the same, making it useful for data masking and cryptographic functions.
5	Encryption	is the process of converting data into a coded format to prevent unauthorized access, ensuring confidentiality and security during storage or transmission.
6	Decryption	is the process of converting encrypted data back into its original, readable format using a key or algorithm to restore access.

7	Plaintext	unencrypted, readable data that can be understood without decryption or special processing
8	ciphertext	encrypted data that is unreadable without decryption using the correct key or algorithm.
9	Cleartext	is data that is stored or transmitted without encryption, making it accessible without decryption.
10	Algorithm	is a step-by-step procedure or set of rules used for computation, including encryption and decryption processes in cybersecurity.
11	A Key	a piece of information, such as a string of characters or a cryptographic value, used to encrypt or decrypt data securely.
12	Mitigation	refers to strategies or actions taken to reduce or prevent security risks, vulnerabilities, or threats
13	hardened	refers to the process of strengthening security measures on a system, network, or device to reduce vulnerabilities and resist attacks.
	Use cases for protecting data using cryptography	
14	confidentiality	is the principle of ensuring that information is accessible only to authorized individuals and protected from unauthorized access.
15	integrity	refers to maintaining the accuracy, consistency, and trustworthiness of data by preventing unauthorized modification or corruption.
16	authentication	is the process of verifying the identity of a user, system, or entity to ensure authorized access
17	nonrepudiation	ensures that an individual or entity cannot deny having performed an action, such as sending a message or making a transaction, through cryptographic proof.

18	Obfuscation	is the practice of deliberately making data, code, or communication more difficult to understand to protect sensitive information from unauthorized access.
19	Security through obscurity	is a security approach that relies on keeping implementation details secret to provide protection, rather than relying on robust security mechanisms.
20	Cryptography Algorithm	is a mathematical procedure used for encrypting and decrypting data to ensure confidentiality, integrity, and security.
21	1. Symmetric algorithm	A cryptographic method that uses the same key for both encryption and decryption.
22	1.1. Private key	A single, secret key used for encryption and decryption in symmetric cryptography.
23	1.1.1. Advanced Encryption Standard (AES)	A widely used symmetric encryption algorithm known for its speed and security.
24	1.1.2. Blowfish	A symmetric block cipher designed for speed and flexibility, using variable-length keys.
25	1.1.3. twofish	A symmetric block cipher, an improvement over Blowfish, offering enhanced security and efficiency.
26	1.1.4. Data encryption standard (DES)	A legacy symmetric encryption algorithm that uses a 56-bit key and operates on 64-bit blocks.
27	1.1.5. Triple Data encryption standard (3DES)	An enhanced version of DES that applies the DES algorithm three times for improved security.
28	1.1.6. Rivest Cipher (RC)	A family of symmetric encryption algorithms, including RC2, RC4 (a popular stream cipher), and RC6
29	2. Asymmetric algorithm (public key cryptography)	cryptographic method that uses a key pair: a public key for encryption and a private key for decryption.
30	2.1. Key pairs	combination of a public key and a private key used in asymmetric encryption.
31	2.2. Public key	key that can be shared openly and is used to encrypt data in asymmetric cryptography.

32	2.3. Private key	secret key that is kept confidential and is used to decrypt data in asymmetric cryptography.
33	2.4. Both direction	Asymmetric encryption allows secure data exchange in both directions, ensuring confidentiality and authentication.
34	2.5. RSA	A widely used asymmetric encryption algorithm that relies on the mathematical difficulty of factoring large prime numbers.
35	2.6. Elliptic curve Cryptography (ECC)	A more efficient asymmetric encryption method that provides strong security with smaller key sizes.
36	2.7. Digital Signature Algorithm (DSA)	cryptographic algorithm used for verifying the authenticity and integrity of digital messages.
37	2.8. Digital signature	cryptographic technique that ensures the authenticity, integrity, and non-repudiation of a message.
38	2.9. Key exchange	method for securely sharing cryptographic keys over an insecure network.
39	2.9.1. Diffie-Hellman (DH)	cryptographic protocol that enables two parties to establish a shared secret key over a public channel.
40	2.9.2. Diffie-Hellman Ephemeral (DHE)	variation of DH that generates temporary keys for each session to enhance security.
41	2.9.3. Elliptic Curve Diffie-Hellman (ECDH)	key exchange protocol based on elliptic curve cryptography, offering stronger security with smaller keys.
42	2.9.4. Perfect Forward secrecy	cryptographic property ensuring that past communication remains secure even if long-term keys are compromised.
43	3. Hashing algorithm	A mathematical function that transforms input data into a fixed-size hash value, ensuring data integrity.
44	One-time pad (OTP)	cryptographic technique that uses a truly random key of the same length as the message for unbreakable encryption.
45	Stream cipher	An encryption method that encrypts data one bit or byte at a time, often used for fast and real-time encryption

46	Block cipher	An encryption method that processes data in fixed-size blocks, commonly used in symmetric encryption.
47	Sponge function	A cryptographic function used in hash algorithms and random number generation, known for absorbing and squeezing data.
48	Hashing	The process of transforming data into a unique fixed-length hash value, used for integrity verification.
49	4. Fixed size	Hash functions produce output of a consistent, predetermined length, regardless of input size.
50	5. Unique (in terms of hashing)	a well-designed hash function ensures that each unique input produces a unique hash output.
51	6. original	cryptographic hash should make it computationally infeasible to derive the original data from the hash value
52	7. secure	secure hashing algorithm resists collisions, preimage attacks, and ensures data integrity.
53	List the family of Message Digest (MD)	A group of cryptographic hash functions, including MD2, MD4, and MD5, though now considered weak.
54	Secure Hash Algorithm (SHA)	family of cryptographic hash functions (SHA-1, SHA-2, SHA-3) designed for data integrity and security.
55	Race integrity primitives evaluation MD (RipeMD)	cryptographic hash function designed for data integrity verification, with variants like RIPEMD-128 and RIPEMD-160.
56	Whirlpool	cryptographic hash function based on a block cipher, designed for security and efficiency.
Encryption through Software (file, disk levels, and across database)		
57	File-level encryption	refers to the process of encrypting individual files on a system, ensuring that each file is protected with its own encryption key

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59	Third party encryption tools	Third-party encryption tools provide software solutions for encrypting files and folders to enhance data security
60	· GNU Privacy Guard (GnuPG)	A free and open-source tool used for file encryption, digital signatures, and key management, offering strong encryption using the OpenPGP standard
61	· AxCrypt	A file encryption software that provides easy-to-use AES-128 and AES-256 encryption for protecting files with strong security.
62	· Folder Lock	A tool that offers encryption for files and folders, along with secure backup, password protection, and data hiding features.
63	· VeraCrypt	A free, open-source disk encryption tool that can create encrypted virtual disks, offering strong security for file and full disk encryption.
Operating System Encryption		
64	Microsoft's Encryption File system	EFS is a feature in Microsoft Windows that provides file-level encryption, allowing users to securely encrypt files and folders on NTFS-formatted drives to protect sensitive data from unauthorized access.
65	Apple's FileVault	FileVault is Apple's full-disk encryption system, available on macOS, that uses XTS-AES-128 encryption with a 256-bit key to secure the entire disk, ensuring that all data is encrypted automatically when the system is in use.
Disk Encryption		
66	Disk-level Encryption	refers to the process of encrypting an entire disk or hard drive, protecting all data stored on the drive, including the operating system, files, and applications, from unauthorized access.

67	Volume-Level Encryption	encrypts specific volumes or partitions of a disk rather than the entire disk, allowing users to protect sensitive data while leaving other data unencrypted for performance or usability reasons.
68	Partition-level encryption	involves encrypting individual partitions of a disk, securing data within a specific partition while leaving other partitions unencrypted
	Database Encryption	
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70	Database-level encryption	involves encrypting an entire database to protect sensitive data from unauthorized access
71	Transparent data encryption (TDE)	a type of database encryption that automatically encrypts the database files at the storage level
72	Record-level encryption	involves encrypting individual records within a database, providing more granular control over which specific data is protected
73	Data/information classification levels	Data/information classification levels define how different types of data should be handled based on their sensitivity and value to an organization.
	Hardware Encryption	
74	USB Devices	are portable storage devices that connect to computers via a USB interface.
75	Self-Encryption Drives (SEDs)	are storage devices, such as USB drives or hard drives, that automatically encrypt the data written to them using built-in encryption algorithms
76	Hardware Security Module (HSM)	is a physical device used to generate, store, and manage cryptographic keys securely
77	Trusted Execution Environment (TEE)	is a secure area within a processor that runs code in isolation from the rest of the system, providing a protected environment

78	Trusted Platform Module (<i>TPM</i>)	a dedicated microprocessor used to securely store cryptographic keys, passwords, and other sensitive data, enabling hardware-based security features such as disk encryption and secure boot.
	Blockchain	
79	Blockchain	is a decentralized, distributed ledger technology that securely records transactions across multiple computers
80	Public blockchain/open public ledger	is a fully decentralized blockchain where anyone can participate, verify transactions, and access the ledger
	Private blockchain	is a permissioned blockchain where access to the network and the ability to validate transactions are restricted to specific entities or organizations
81	Federal blockchain or consortium blockchain	a hybrid model where multiple organizations or entities share control over the blockchain network.
	Attacks on Cryptography	
82	Algorithm attacks	target weaknesses in cryptographic algorithms, aiming to exploit vulnerabilities in the encryption methods
83	Known ciphertext attacks	occur when an attacker has access to a ciphertext (encrypted message) and attempts to deduce the corresponding plaintext or key, using cryptanalysis techniques.
84	Downgrade attacks	forces a system to fall back to a weaker version of a protocol or encryption method
85	Attacks based on Misconfiguration	exploit incorrect or inadequate security settings, such as weak passwords, improper access controls, or unsecured default configurations, to gain unauthorized access to systems or data.
86	Collision attacks (in terms of hashing)	on hashing algorithms attempt to find two different inputs that produce the same hash value

87	Birthday attack	is a cryptographic attack that leverages the probability of finding two distinct inputs that hash to the same value
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